The alternative usually taken is to discard the hydraulics and command joint positions directly, which removes the load and pressure dependence that makes the machine difficult to control. What is needed is a representation that is cheap enough to train against and faithful enough that the resulting policy is controlling something close to the real actuator

Learned actuator models and latent-conditioned adaptation are both established techniques and Section 1.2 described the progress that makes them practical on machines of this class. What remains unclear is how such models should be identified so that they retain the command dependence a controller needs, whether they transfer between simulators without being retrained, and whether online adaptation adds anything once a policy has been trained against them. This thesis therefore treats representation and transfer as the primary questions, and treats online adaptation as the last of them rather than the first The research statement of the thesis is as follows:

Comparison notation.

For readability, the three policy–actuator stacks are denoted by short labels throughout this thesis: *R* is PPO–ROM, *IF* is PPO–ID–FD, and *AIF* is PPO–(AAM)–ID–FD. R uses the command-conditioned reduced-order model; IF uses the frozen inverse/forward pair and AIF augments IF with online adaptation. The labels name the complete stacks, not just their actuator models.

> *To determine how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, and under what conditions online adaptation improves on the robustness those learned models already provide.*

The statement decomposes into six research questions, which are answered in the order given. The first three concern how the actuator is represented, the fourth concerns control through that representation, and the last two concern what remains for adaptation once the first four are settled. The six questions are:

1. How can data from a high-fidelity Simscape model be used to learn computationally tractable representations of hydraulic actuation while preserving the command dependence a controller requires?
2. To what extent do actuator models identified from Simscape transfer to an IsaacLab manipulator without deployment-specific retraining?
3. Can a physics-structured reduced-order hydraulic model supply dynamics that are accurate and stable enough for GPU-parallel policy training?
4. Can a policy trained from a privileged classical teacher become the sole deployed controller while acting through learned actuator dynamics?
5. How does the choice of actuator representation affect tracking, command behaviour latency and robustness under plant variation?